Proposal presentation script

Qihang Feng | ECE 730 | About five minutes

Read the paragraphs aloud. Slide headings and timing notes are for rehearsal only.

Slide 1 Effective prediction utilization

About 35 seconds

Hello, my name is Qihang Feng. My project explores how to make better use of a robot model's own predictions. I will use PushT, a two-dimensional simulator where a pusher moves a T-shaped block onto a target. Camera feedback connects physical motion with software decisions. I will study two approaches: adaptive action execution, and fine-tuning with successful trajectories generated by the model itself.

Slide 2 Prediction and execution

About 50 seconds

I will build on a pretrained Diffusion Policy, which predicts a sequence of movements from camera images and the pusher's position. The original baseline executes eight movements, then uses updated observations to predict again. This fixed interval provides regular feedback, but some unused movements may still be useful.

My controller will check whether to replan after eight movements or continue for up to four more. The model will generate the same sequence length with the same inference settings. Only the decision about how much of that sequence to execute will change.

Slide 3 Signals for continuing or replanning

About 65 seconds

The controller will consider estimated target alignment, recent changes in alignment, and the smoothness of the remaining movement path. Alignment means how closely the block covers the target.

If the visual estimate is clear, recent decline is limited, and the remaining path is smooth, the robot may continue to twelve movements. Near the target, or when these checks fail, it will replan after eight.

During execution, repeated warnings and accumulated alignment loss may also trigger an early interruption. However, a temporary drop does not automatically mean failure. The robot may need to reposition before making progress. I will look for signs of recovery and limit how often interruptions occur.

Slide 4 Self imitation based fine tuning

About 55 seconds

The second approach is self-imitation-based fine-tuning. In the diagram, I will return to a saved intermediate state from a failed baseline attempt and sample several continuations using the original model. The retries start from that saved state, not from the failed endpoint. Because the model is stochastic, different predictions may produce different outcomes.

I will keep trajectories that eventually finish the task and mix them with expert demonstrations. These examples will fine-tune only the model's final convolution block, while the other parameters stay frozen. The updated model will then use the same adaptive controller. Its parameters will remain fixed during evaluation.

Slide 5 Planned evaluation

About 55 seconds

I will compare the four configurations shown here: the original and fine-tuned models, each with fixed-eight and adaptive execution. This will help separate the effects of the two approaches.

The comparisons will use matching initial scenes, repeated prediction samples, and equal movement limits. Fine-tuning and evaluation scenes will remain separate.

I will measure success rate, average final alignment, average prediction calls, and average total runtime. Success requires more than ninety-five percent target coverage. Runtime will include monitoring, and failed attempts will remain in the averages. I will also check which cases improve and which become worse.

Slide 6 Expected contribution and scope

About 35 seconds

The goal is more successful pushes with fewer predictions and shorter runtime. I will also evaluate final alignment and report uncertainty, rather than assume that every metric will improve.

The experiments will use two-dimensional simulation on my GTX 1650 laptop. Simulation pauses while the model predicts movements, so this study will not demonstrate real-time control or performance on a physical robot.

Slide 7 Thank you

About 5 seconds

Thank you for listening. I'm happy to take your questions.